



USN

NITTE (Deemed to be University)

(Established under Sec.(3) of UGC Act, 1956)

First Semester M.Tech (CBCS) Degree Examinations, December 2025

25CYS133 – CLOUD SECURITY AND IOT SECURITY

(For CYS)

Duration: 3 Hours]

[Max. Marks: 100

Note: Answer **Five full** questions choosing **Two full** questions from **Module – I & Module – II** each and **One full** question from **Module – III**.

		Module – I			
		Marks	BT*	CO*	PO*
1.	a) Explain various Cloud Deployment Models with examples.	10	2	1	1
	b) Discuss the Shared Responsibility Model for IaaS, PaaS, and SaaS environments.	10	2	1	1
2.	a) Describe the Cloud Delivery and Deployment Architecture. Explain how virtualization supports cloud operations.	10	2	2	3
	b) Explain the Threat Model of a Cloud Architecture and discuss major cloud threats.	10	2	1	1
3.	a) Discuss Data Asset Classification using the CIA Triad in a cloud environment.	10	2	2	3
	b) Explain Regulatory, Compliance, and Legal aspects in cloud data management.	10	2	1	1
		Module – II			
4.	a) Compare Tokenization and Encryption techniques used for protecting cloud data.	10	3	3	2
	b) Explain the process and importance of Key Management in cloud environments.	10	2	3	2
5.	a) Define and explain the components of Identity and Access Management (IAM) in cloud security.	10	2	3	2
	b) Discuss the methodology of Vulnerability Analysis and Penetration Testing in cloud.	10	2	3	2
6.	a) Explain the CSA Security Framework and its relevance to cloud computing.	10	2	3	2
	b) Illustrate the Tools, Techniques, and Procedures (TTPs) used for securing cloud platforms.	10	2	3	2
		Module – III			
7.	a) Explain the IoT Reference Model and describe its Functional View.	10	2	4	3
	b) Discuss the IoT Security Challenges related to hardware and software components.	10	2	4	3
8.	a) Describe various IoT Attacks.	10	2	5	2
	b) Explain IoT Security Requirements such as confidentiality, secure firmware, and trusted analytics.	10	2	5	2

BT* Bloom's Taxonomy; L* Level; CO* Course Outcome; PO* Program Outcome
